

Capstone_Report_Team_08_2

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ABSTRACT

Road accidents remain a major problem and many of them occur due to human-related factors such as drowsiness, distraction, and rash driving. Even a moment of reduced attention can create a dangerous situation, especially during long or late-night journeys. Since these behaviours are hard to notice without continuous monitoring, driver-assistance systems have become important in improving road safety.

This project, “**Driver Behaviour Anomaly Detection using Deep Learning for Accident Prevention in Intelligent Transportation Systems (ITS)**”, aims to build a system that can observe drivers in real time and identify unsafe behaviour. By using deep learning and computer vision, the system analyses video input to detect signs like eye closure, unusual head movement, or looking away from the road. When abnormal behaviour is detected, the system provides an early warning so the driver can correct their action and avoid potential accidents.

During Phase-1, we focused on studying the problem, reviewing previous research, identifying the gaps in existing systems, and designing the overall architecture for our model. We also explored available datasets and examined the challenges involved in building a reliable real-time detection system. The work completed in this phase forms the foundation for Phase-2, where we will implement, train, and evaluate the deep learning model.

CHAPTER 1

INTRODUCTION

Road accidents remain one of the major concerns in transportation systems across the world. A large proportion of these accidents occur because of certain human errors, such as drowsiness, distraction, inattention, over-speeding, and other unsafe driving behaviors. In view of increasing numbers of vehicles on the road and ever-increasing accident cases, there is an urgent need for intelligent systems that can assist drivers to reduce the possibility of accidents .

The "Driver Behavior Anomaly Detection using Deep Learning for Accident Prevention in ITS" project focuses on unsafe driver behavior detection in real time by analyzing the pattern of a driver's face and behavior . This system can detect signs such as eye closure, frequent blinking, head tilting, distraction, or sudden irregular movements with the use of deep learning techniques. Once any abnormal behavior is observed, the system generates an early warning so that the accident can be avoided before it occurs.

Traditional vehicle safety systems mostly operate either after an accident has occurred or with a major emphasis on factors around the vehicle. But if drivers can be warned at the appropriate time, many severe accidents could be avoided. Integrating deep learning-based monitoring systems into Intelligent Transportation Systems ,which are becoming common, will make transportation safer, smarter, and more reliable.

The main focus of the Phase-I work is to understand the nature of unsafe driver behavior, review the existing solutions, identify research gaps, and design a suitable system architecture. The proposed solution will help in reducing accidents by acting as a preventive safety layer that supports the driver through timely alerts.

1.1 Driver Behaviour in Intelligent Transportation Systems (ITS)

Intelligent Transport Systems seek to achieve road safety, decongest roads, and provide a better driving experience with the integration of modern technologies like sensors, communication systems, and artificial intelligence. One important component is driver behavior monitoring within ITS, since the driver plays a major role in road safety.

The risk of accidents increases with unsafe driving actions like drowsiness, distraction, the use of any mobile device, or sudden lane changes. Monitoring such behavior is vital because even the slightest moment of distraction could lead to hazardous situations during long driving or heavy traffic. ITS-based driver monitoring systems observe the drivers face, eyes, and head movements with cameras and other sensors to make out whether the driver is attentive or showing signs of unsafe driving behavior .

ITS can help a vehicle become more aware of the condition of the driver by integrating behavior analysis, thus supporting the driver with timely alerts that will make transportation safer and more reliable.



1.1.1 Need for Real-Time Anomaly Detection

Most driver safety systems rely on either post-accident responses or simple sensors that cannot accurately detect changes in behavior. Human behavior can change in a very short time—drivers can fall asleep, get distracted, or become inattentive in just a few seconds. Early warning signs are usually missed without real-time detection.

Real-time anomaly detection is of essence since it:

- Continuously monitors the drivers eye movements, facial expressions, and head position
- Identifies early signs of fatigue or distraction before they become dangerous
- Immediately warns the driver in case of abnormal behavior detected by it
- Reduces reaction time, enabling the driver to regain control

- Helps prevent accidents, especially during night driving or long-distance journeys Real-time anomaly detection using deep learning makes this system capable of dealing with complex behavioral patterns, thus detecting unsafe actions with much better accuracy . It increases driving safety and complements broader objectives related to the Intelligent Transportation Systems.

CHAPTER 2

PROBLEM STATEMENT

Road safety is still one of the biggest concerns in the modern transportation environment, and most road accidents are caused by human factors. Drivers may exhibit unsafe behaviors in the form of drowsiness, distraction, mobile phone usage, delayed reactions, or sudden loss of attention . These changes in behavior happen suddenly, with the pronounced symptom usually hard to be noticed without continuous monitoring . Therefore, many opportunities for early warnings are lost, and accidents that could have been prevented take place.

Although several vehicle safety technologies exist, most are reactive in nature — that is , they analyze data safety mechanisms only after the driver has already entered into a risky situation. Many systems also rely on basic sensors or rule-based approaches, which cannot understand complex human behavioral patterns. These limitations make traditional methods insufficient for real-time monitoring of the driver's state.

There is a definite need for an advanced, intelligent system that applies deep learning and computer vision techniques for the analysis of live video data of the driver, understanding the behavioral cues, and accurate detection of anomalies. A system capable of detecting early signs of fatigue, distraction , or rash behavior can go a long way in reducing accident risks significantly. The subject problem is essentially related to the need for an appropriate and effective real-time anomaly detection mechanism for driver behavior monitoring as part of ITS.

LITERATURE SURVEY

Details of Paper- Title, author, conference /Journal	Methodology Used	Result	Limitations
¹² Abnormal Driving Behavior Detection : A Machine and Deep Learning Based Hybrid Model (Md. Ashraf Uddin et al., Int. J. Intelligent Transportation Systems Research, 2025)	Hybrid model combining Machine Learning (Random Forest, SVM, Gradient Boosting) on tabular driving data and Deep Learning (CNN, ResNet50, EfficientNetB6) on image data, uses UAH-DriveSet + SFD3 datasets.	Many ML models achieved high accuracy (up to ~96% using boosting / LSTM). Systems could classify behaviours like aggressive, tired, distracted, and normal driving.	Most methods depend on sensor data, not camera input. No unified approach for driver behaviour types. Many datasets are limited or imbalanced.
⁹ Real-Time Driver Fatigue Detection System with Deep Learning on a Low-Cost Embedded System (Esra Civik & Ugur Yuzgec,	Designed two CNN models to detect fatigue using eye and mouth regions. Used dlib for facial landmark extraction. Tested system on a	Achieved ~94% accuracy ²⁰ on the YawDD dataset. The system could run in real time (around 6 fps) even	Processing speed is limited on low-cost devices. Limited to fatigue detection only. Small dataset.

Microprocessors and Microsystems, 2023)	Jetson Nano embedded device.	on low-cost hardware.	
⁵ Comprehensive Study of Driver Behavior Monitoring Systems Using Computer Vision and Machine Learning Techniques (Fangming Qu et al., Journal of Big Data, 2024)	A survey of many AI and ML methods such as CNNs, RNNs, hand-crafted features ,and hybrid systems. The paper reviews facial, hand,and posture-based driver monitoring approaches and compares available datasets.	The study lists different techniques used in driver monitoring and highlights which approaches work best for behaviour analysis. It also identifies hybrid CV + ML systems as promising.	The work is mostly theoretical. It doesn't include real implementation or quantitative testing.
¹⁰ Deep Learning Model for Driver Behavior Detection in Cyber-Physical System-Based Intelligent Transport Systems (Brij B. Gupta et al., IEEE Access, 2024)	¹⁸ Developed a CNN-based model for detecting driver behaviour inside ITS. Used data augmentation and transfer learning to handle limited labelled data. Integrated with CPS architecture.	The model achieved good accuracy for classifying eye states, yawning ,and fatigue patterns. It showed potential for real-time use in ITS.	Needs large labelled datasets. Training is computationally heavy ,and real-time performance depends on hardware.
Naturalistic Driving Data-Based Anomalous Driving Behavior Detection Using Hypertuned	Built a deep autoencoder to learn normal driving behaviour from naturalistic vehicle	Detected unsafe behaviours such as sudden braking ,lane changes ,or irregular	Works only on trajectory data ,not video or facial behaviour. Does not classify the type of

Deep Autoencoders (Shafqat Abbas et al., Electronics, 2023)	trajectory data (NGSIM dataset). Used reconstruction - error - based anomaly scoring.	movement. Accuracy was around 95–96 % based on anomaly score thresholds.	unsafe behaviour. Requires fine-tuning of thresholds for different environments.
4 An Enhanced Deep Neural Network Model for Detection of Anomalous Behavior of Drivers in Road Traffic (2024 ,Wiley Journal of Surveillance)	Built an enhanced CNN model using transfer learning (ResNet50, VGG19, MobileNet). Used YOLOv3 for vehicle detection and additional layers for multiclass anomaly detection (overspeeding, no-stopping-zone violations ,spacing issues, etc.).	Final system achieved around 95% accuracy across four anomaly types. Showed strong improvement over baseline CNN models.	Needs large annotated datasets for training. The model mainly works with video surveillance data ,not in-cabin driver monitoring.

RESEARCH GAP

Most of the driver monitoring systems in existence monitor only one aspect, either drowsiness or simple distraction, and usually depend on basic sensors that are not very reliable. Lighting, camera angle, and driver movement are subject to great variation in natural driving conditions. Most systems do not perform well under these changes. Another limitation is that several deep learning models used in previous works were trained using small datasets; thus, they do not generalize well when they are tested in new environments.

There is also limited research on lightweight models that can run smoothly on low-cost or embedded hardware, which is essential for practical in-vehicle systems. Very few studies combine both the visual cue and sensor information for a more accurate understanding of the driver's condition. Furthermore, many existing approaches cannot describe why a certain behavior is flagged as unsafe, which makes the system less transparent.

These gaps illustrate a need for a more robust, adaptable, and real-time driver behavior detection system that should work in varying conditions and be able to provide reliable feedback to the driver.

CHAPTER 5

OBJECTIVES

- Identifying unsafe driver behavior, such as drowsiness, distraction, and rash driving
- Using deep learning, build a system that could conduct real-time driver video analysis .
- Reducing accidents by the provision of early warnings in case abnormal behavior is detected .
- Lightweight model design: suitable to run on in-vehicle hardware.
- Proper pre-processing and tuning of the models, so that better accuracy with reduced false alarms can be achieved .
- Developing an adaptive monitoring system for different drivers and different environments.
- Support ITS to reduce accidents caused by human error.

Overview of Datasets

6.1 Description :

The dataset used for the next project is called ¹⁷the State Farm Distracted Driver Detection Dataset. This is a real-world example of a computer vision dataset extracted from a competition in Kaggle. The data consists of images of drivers engaging in various activities, both unsafe and safe, taken from dashboard cameras. It finds wide application in the deep learning study of driver monitoring systems because it performs multi-class behavior classification.

⁵The dataset comprises 10 categories ranging from c0–c9, including typical distracted behaviors like texting, talking on the phone, operating the radio, drinking, and reaching behind. This categorization thus helps in modeling different unsafe driver actions by presenting the deep learning model with helpful visual information to detect anomalies in real time.

This dataset is suitable for the project because it:

- This includes real in-car camera images matching the goal of the project.
- Large and diverse, there is variation of lighting, angle, and appearance of the driver.
- This also challenges the model with imbalanced classes, requiring robust training and data augmentation strategies.

It would thus provide a sound basis for the driver behavior anomaly detection system using CNN-based architectures.

6.2 Data Attributes / Features

The components in this dataset include :

A. Image Classes

CLASS	DESCRIPTION
C0	Normal driving
C1	Texting-right hand
C2	Taking on phone –right hand
C3	Texting-left hand
C4	Taking on phone-left hand
C5	Operating the radio
C6	Drinking
C7	Reaching behind
C8	Hair and makeup
C9	Taking to passenger

B. Dataset Organization :

state-farm-distracted-driver-detection/

| |-----imgs/

| | |-----train/

| | | |-----C0/ ... C9/

| | |-----test/

| | | |-----test_stg2/

|

| |-----driver_imgs_list.csv

| |----- sample_submission.csv

C. Dataset Statistics

FEATURE	DESCRIPTION
Total images	~10,000
Train images	~22,424

Test images	~79,000
Image size	variable
Classes	10
Drivers	26 unique driver Id's
Type	RGB
Format	JPEG(.jpg)

D. Challenges in the Dataset

Class imbalance

Different lighting conditions

- Driver appearance variation
- Reflections and shadows
- Some behaviors look visually similar (for example, c1 versus c3)

These challenges make this dataset appropriate for evaluations concerning the robustness of deep learning models.

CONCLUSION OF CAPSTONE PROJECT PHASE – 1

In Phase–1 of the project, we focused on understanding the problem of unsafe driver behaviour and the growing need for real-time monitoring systems in Intelligent Transportation Systems. Human factors such as drowsiness, distraction and lack of attention continue to be major reasons for road accidents ,and this provided the motivation for studying different approaches used in driver behaviour detection. In this phase, we went through several research papers to understand various existing techniques, datasets, and models used for unsafe behavior detection. The literature study revealed that deep learning-based models, in particular the CNN-based approaches, perform well in analyzing facial cues, eye state, or head movements. We also found that most of the existing studies suffer from issues such as small datasets, low performance in real-world conditions, and lack of support for multiple behavioral types to be detected together.

We identified the research gap and defined key objectives for our work based on this study, after which an analysis of publicly available datasets and the features needed for behaviour detection was carried out. Using those insights, we prepared the overall system architecture for our proposed solution.

In short, the work done in Phase–1 has established ground for the following phase. It helped us clearly understand the problem, analyze existing solutions, identify gaps, and also plan a direction for the implementation phase . Phase–2 will focus on model development, dataset preparation, training, testing, and performance evaluation to build a reliable real-time driver behavior anomaly detection system.

PLAN OF WORK FOR CAPSTONE PROJECT PHASE – 2

Actual development of the driver behavior anomaly detection system will be carried out at Phase 2, based on the design and understanding developed during Phase 1. This includes dataset preparation, model development, testing, and performance evaluation. Following tasks are planned during the course of Phase 2:

1. Data Collection & Preprocessing:

Identify driver monitoring datasets that have Drowsy/Distracted/Normal Behaviour features.

- Extract video data frames and label them using behavioural categories.

Preprocessing steps shall include at least the following: resizing, normalizing inputs, extracting the face or eye region, and augmentation of data.

2. Model Development :

This will be achieved with deep learning modeling using CNN or a hybrid model helpful in improving accuracy.

This model can be trained according to one's needs for either the detection of eye state, distraction detection, or head movement.

- Hyperparameter tuning to achieve high accuracy with a minimum number of false detections.

3. System Integration :

Summary: In the architecture described below, all detection modules are incorporated in a single pipeline.

Input can either be provided by a webcam or from a video stream to realize real-time behavioral monitoring.

- Design an alert mechanism that warns in case of detected abnormal behavior.

4. Testing and Performance Evaluation:



- Real-time video input will be provided for testing accuracy, speed, and reliability.
- Quantify the performance across illumination changes, camera position, and driver variation. It is to be compared with existing methods to find the improvement.

5. Optimization for Deployment:

- If required, reduce the model size to make it real time.
- Increase the frame processing speed to fit real time constraints.
- Make sure the system runs with no significant computation using standard hardware.

6. Documentation and Final Report:

- Performance of the model, challenges, and implementation details will be documented.
- Final project report and presentation will be prepared for evaluation.

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